PSY 5790: Applied Univariate and Bivariate/Bivariable Statistics

Continuous Data Analysis Demo 2 — ANOVA

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```
# install.packages("datarium")
# install.packages("rstatix")
# install.packages("ggpubr")

library(jmv)
library(psych)
library(tidyverse)
library(infer)
library(datarium)
library(rstatix)
library(ggpubr)
```

Repeated-Measures (Within-Subject) ANOVA

Single DV collected multiple times from Ss

Research Question 1: Are the self-esteem means different across the three time points for our participants?

```
# Load and inspect demo data set(s)

# Load `mice` data from `datarium` package
data(selfesteem, package="datarium")
head(selfesteem)
```

```
## # A tibble: 6 × 4
##      id    t1    t2    t3
##   <int> <dbl> <dbl> <dbl>
## 1     1  4.01  5.18  7.11
## 2     2  2.56  6.91  6.31
## 3     3  3.24  4.44  9.78
## 4     4  3.42  4.71  8.35
## 5     5  2.87  3.91  6.46
## 6     6  2.05  5.34  6.65
```

Hypothesis Statement: 2-tailed (Non-directional)

H_0: Null Hypothesis — Self-esteem means are not different across time, $\mu_{Time1} = \mu_{Time2} = \mu_{Time3}$

H_a: Alternate Hypothesis — At least one mean (μ) differs from another

Descriptive Statistics (Do not forget to visualize!!)

First, we will change the format from wide to long!

```
selfesteemB <- selfesteem %>%
  gather(key = "time", value = "selfest", t1, t2, t3) %>%
  convert_as_factor(id, time)
head(selfesteemB, 3)
```

```
## # A tibble: 3 × 3
##   id    time  selfest
##   <fct> <fct>   <dbl>
## 1 1     t1       4.01
## 2 2     t1       2.56
## 3 3     t1       3.24
```

```
#Descriptive statistics for `self-esteem` variable across `time`
describeBy(selfesteemB$selfest, group=selfesteemB$time, mat=TRUE)
```

```
##     item group1 vars  n     mean        sd   median  trimmed       mad      min
## X11    1     t1    1 10 3.140122 0.5518061 3.211833 3.168791 0.4524076 2.045868
## X12    2     t2    1 10 4.933942 0.8630697 4.600536 4.814760 0.6019745 3.908429
## X13    3     t3    1 10 7.636342 1.1425976 7.463227 7.534572 1.4009361 6.308434
##          max    range       skew    kurtosis        se
## X11 4.005027 1.959159 -0.4466668 -0.67017272 0.1744964
## X12 6.912915 3.004487  1.0074034  0.05150281 0.2729266
## X13 9.778410 3.469977  0.4046384 -1.29566568 0.3613211
```

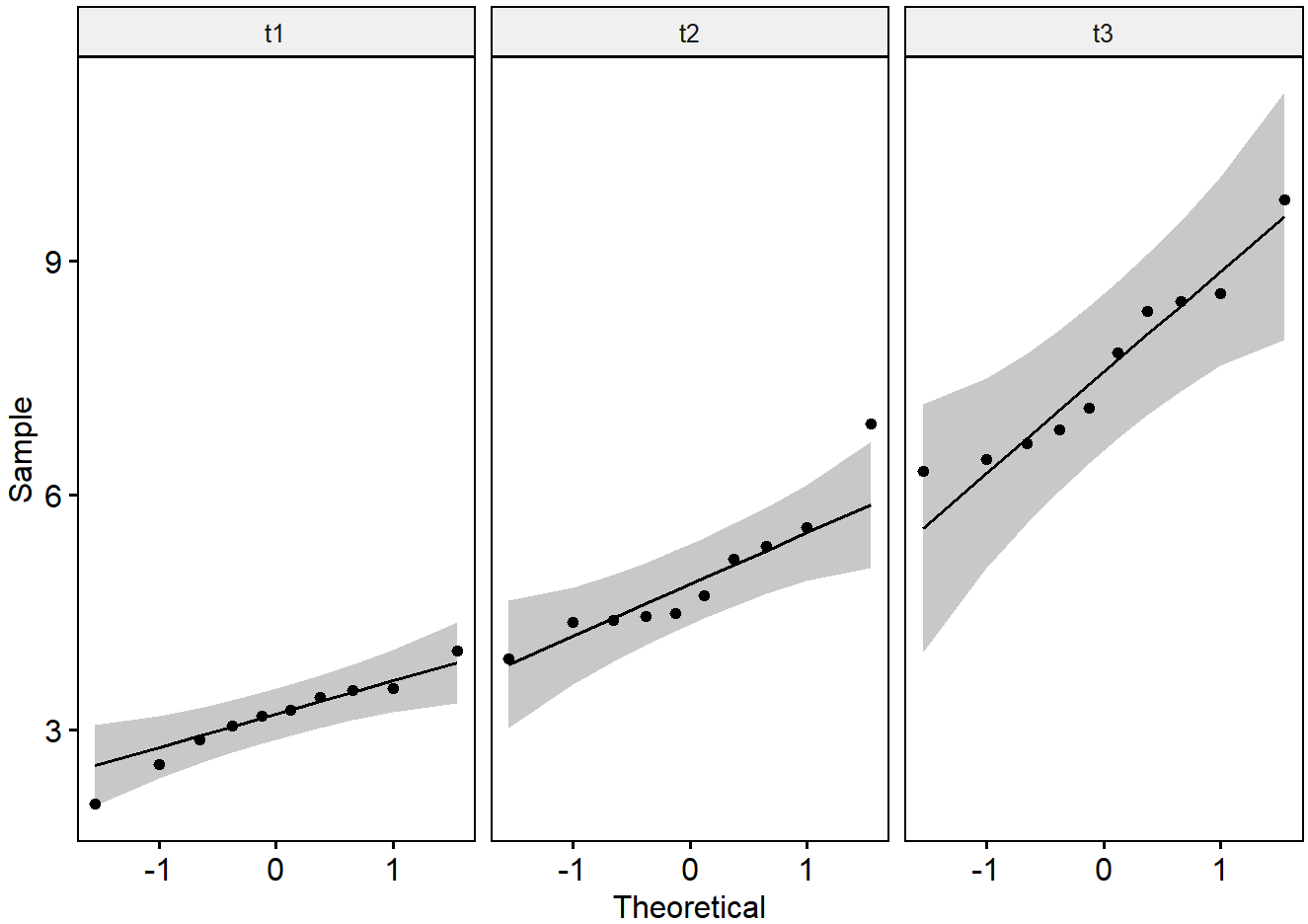
`jmv` Repeated Measures ANOVA

Assumption Checks

```
# Normality of DV across levels of IV tests

selfesteemB %>%
  group_by(time) %>%
  shapiro_test(selfest)
```

```
## # A tibble: 3 × 4
##   time  variable statistic     p
##   <fct> <chr>        <dbl> <dbl>
## 1 t1    selfest      0.967 0.859
## 2 t2    selfest      0.876 0.117
## 3 t3    selfest      0.923 0.380
```

```
ggqqplot(selfesteemB,
         "selfest",
         facet.by = "time")
```




```
# Sphericity check for when 3 + RM values used!

selfesteem %>%
  anovaRM(rm = list(
    list(label = 'Timing',
         levels = c('Time1','Time2','Time3'))),
    rmCells = list(
      list(measure = 't1',
           cell = 'Time1'),
      list(measure = 't2',
           cell = 'Time2'),
      list(measure = 't3',
           cell = 'Time3')),
    spherTests = TRUE) #This is the only information we really need to interpret AT THIS POINT!!
```

```
##
##  REPEATED MEASURES ANOVA
##
##  Within Subjects Effects
##  ─────────────────────────────────────────────────────────────────────────────
##                 Sum of Squares    df    Mean Square    F           p
##  ─────────────────────────────────────────────────────────────────────────────
##    Timing           102.45582     2     51.2279125    55.46903    < .0000001
##    Residual          16.62374    18      0.9235409
##  ─────────────────────────────────────────────────────────────────────────────
##    Note. Type 3 Sums of Squares
##
##
##  Between Subjects Effects
##  ─────────────────────────────────────────────────────────────────────
##                 Sum of Squares    df    Mean Square    F    p
##  ─────────────────────────────────────────────────────────────────────
##    Residual          4.570442     9      0.5078269
##  ─────────────────────────────────────────────────────────────────────
##    Note. Type 3 Sums of Squares
##
##
##  ASSUMPTIONS
##
##  Tests of Sphericity
##  ──────────────────────────────────────────────────────────────────────────────────
##               Mauchly's W    p            Greenhouse-Geisser ε    Huynh-Feldt ε
##  ──────────────────────────────────────────────────────────────────────────────────
##    Timing      0.5508534    0.0920755               0.6900613        0.7743711
##  ──────────────────────────────────────────────────────────────────────────────────
```

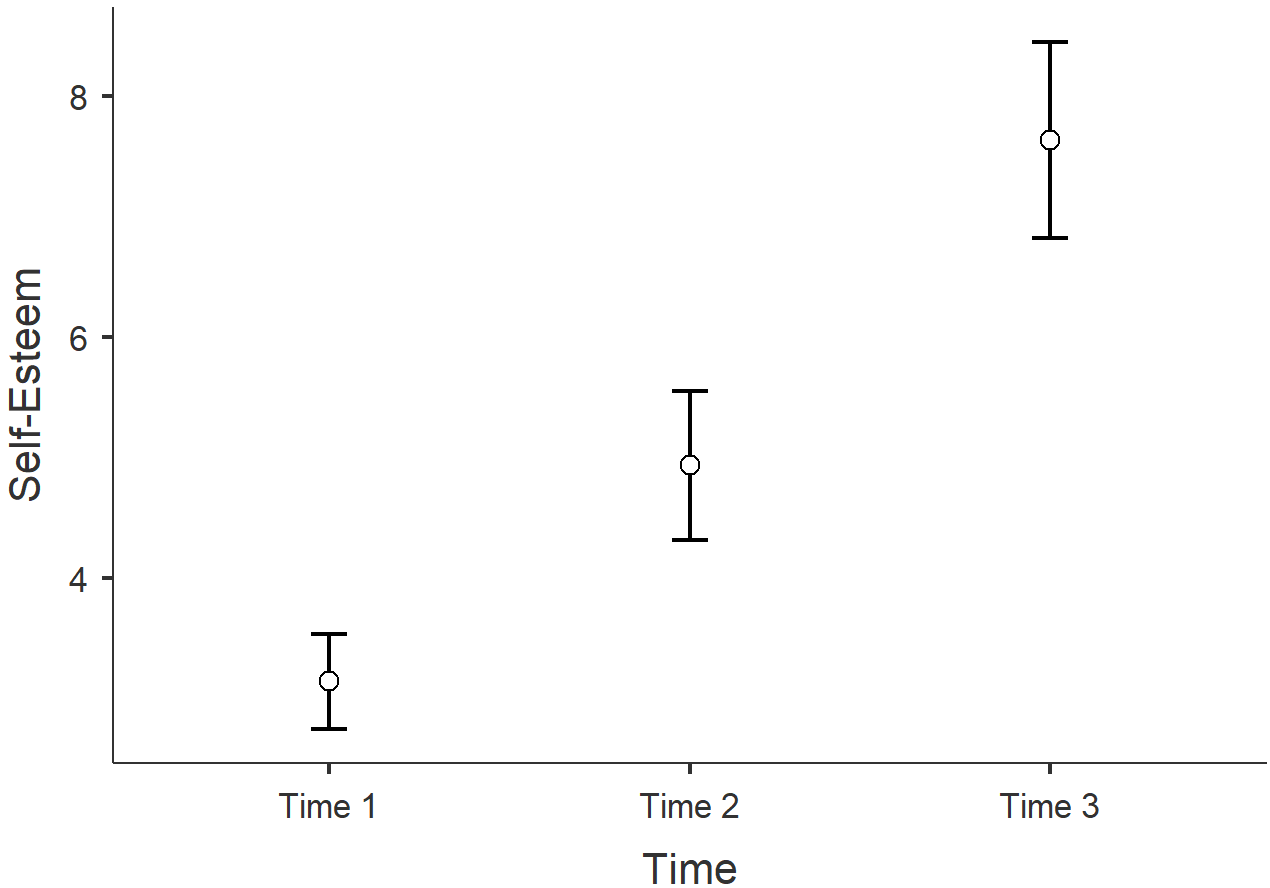
Running the RM ANOVA: `jmv`

```
selfesteem %>%
  anovaRM(rm = list(list(
    label = 'Time',
    levels = c('Time 1','Time 2','Time 3'))),
         rmCells = list(
           list(
             measure = 't1',
             cell = 'Time 1'),
           list(
             measure = 't2',
             cell = 'Time 2'),
           list(
             measure = 't3',
             cell = 'Time 3')),
    effectSize = 'eta',
    spherCorr = list(c('none','GG','HF')),
    spherTests = TRUE,
    postHoc = 'Time',
    postHocCorr = 'bonf',
    depLabel = "Self-Esteem",
    emMeans = list(list("Time")),
    emmPlots = TRUE,
    groupSumm = TRUE)
```

```
##
##  REPEATED MEASURES ANOVA
##
##  Within Subjects Effects
##  ──────────────
```

```
──────
##    Time 1    -    Time 2          -1.793820    0.3611027    9.000000     -4.967618       0.00
23173
##              -    Time 3          -4.496220    0.3398979    9.000000    -13.228148       0.00
00010
##    Time 2    -    Time 3          -2.702400    0.5551565    9.000000     -4.867816       0.00
26586
##  ──────────────────────────────────────────────────────────────────────────────────────────────
──────
##
##
##  Group Summary
##  ─────────────────────
##         N     Excluded
##  ─────────────────────
##         10           0
##  ─────────────────────
```

```
## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## ℹ Please use the `linewidth` argument instead.
## ℹ The deprecated feature was likely used in the jmvcore package.
##   Please report the issue at <https://github.com/jamovi/jmvcore/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

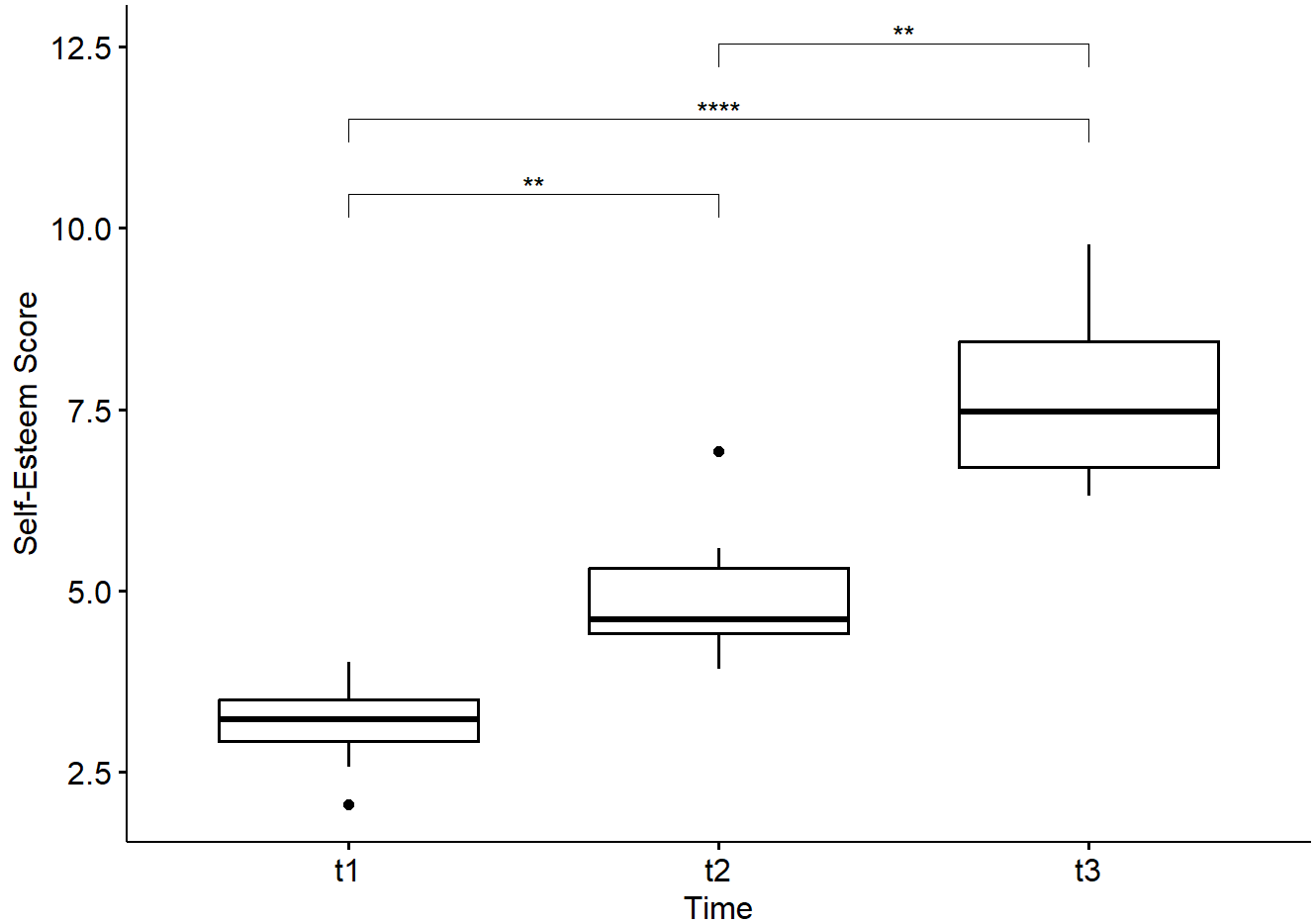


Publishable Plot | Using `rstatix` and `ggpubr` packages

```
# Generate pairwise comparison table with xy positioning

pwc <- selfesteemB %>%
  pairwise_t_test(selfest ~ time,
                  paired = TRUE,
                  p.adjust.method = "bonferroni") %>%
  add_xy_position(x = "time")

pwc
```

```
## # A tibble: 3 × 14
##   .y.     group1 group2    n1    n2 statistic    df         p p.adj p.adj.signif
##   <chr>   <chr>  <chr>  <int> <int>     <dbl> <dbl>     <dbl> <dbl> <chr>
## 1 selfest t1     t2        10    10     -4.97     9   7.72e-4  2e-3 **
## 2 selfest t1     t3        10    10    -13.2      9   3.34e-7  1e-6 ****
## 3 selfest t2     t3        10    10     -4.87     9   8.86e-4  3e-3 **
## # ℹ 4 more variables: y.position <dbl>, groups <named list>, xmin <dbl>,
## #   xmax <dbl>
```

```
# Generate publishable plot from long data

bxp <- ggboxplot(selfesteemB,
                 x = "time",
                 y = "selfest") +
  labs(x = "Time",
       y = "Self-Esteem Score") +
  stat_pvalue_manual(pwc, hide.ns = TRUE)
bxp
```



APA Style Results Section

Statistical Interpretation:

A within-subject one-way ANOVA was performed to determine the relationship between self-esteem scores and timing. The mean and standard deviations are presented in Table 1. The ANOVA was statistically significant at the .05 alpha level, $F(2, 18) = 55.47$, $p < .001$, $\eta^2 = .83$. A post hoc test was conducted using the Bonferroni Correction that indicated that the mean self-esteem level at Baseline were statistically significantly lower than self-esteem levels at Time 1 ($p = .002$) and at Time 2 ($p < .001$). Furthermore the mean self-esteem level at Time 1 was statistically significantly lower than self-esteem levels at Time 2 ($p = .003$).

One-Way (Between-Subjects) ANOVA

Single DV collected across **mutually exclusive** groups (3+)

Research Question 1: Do different nutrient treatments (e.g., `control`, `calcium`, `nitrogen`) lead to different levels of `plant growth` (i.e., `weight`)?

```
# Load and inspect demo data set(s)

# Load `mice` data from `datarium` package
data("PlantGrowth")
head(PlantGrowth)
```

```
##   weight group
## 1   4.17  ctrl
## 2   5.58  ctrl
## 3   5.18  ctrl
## 4   6.11  ctrl
## 5   4.50  ctrl
## 6   4.61  ctrl
```

Hypothesis Statement: 2-tailed (Non-directional)

H_0: Null Hypothesis — Plant growth (i.e., `weight`) means are not different across nutrient treatments, $\mu_{Control} = \mu_{Calcium} = \mu_{Nitrogen}$

H_a: Alternate Hypothesis — At least one mean (μ) differs from another

Descriptive Statistics (Do not forget to visualize!!)

```
#Descriptive statistics for `weight` variable across `group`
describeBy(PlantGrowth$weight, group=PlantGrowth$group, mat=TRUE)
```

```
##     item group1 vars  n  mean        sd median trimmed      mad  min  max range
## X11    1   ctrl    1 10 5.032 0.5830914  5.155 5.00500 0.719061 4.17 6.11  1.94
## X12    2   trt1    1 10 4.661 0.7936757  4.550 4.62375 0.533736 3.59 6.03  2.44
## X13    3   trt2    1 10 5.526 0.4425733  5.435 5.50375 0.363237 4.92 6.31  1.39
##          skew  kurtosis        se
## X11 0.2311020 -1.116799 0.1843897
## X12 0.4744580 -1.104735 0.2509823
## X13 0.4847153 -1.160361 0.1399540
```

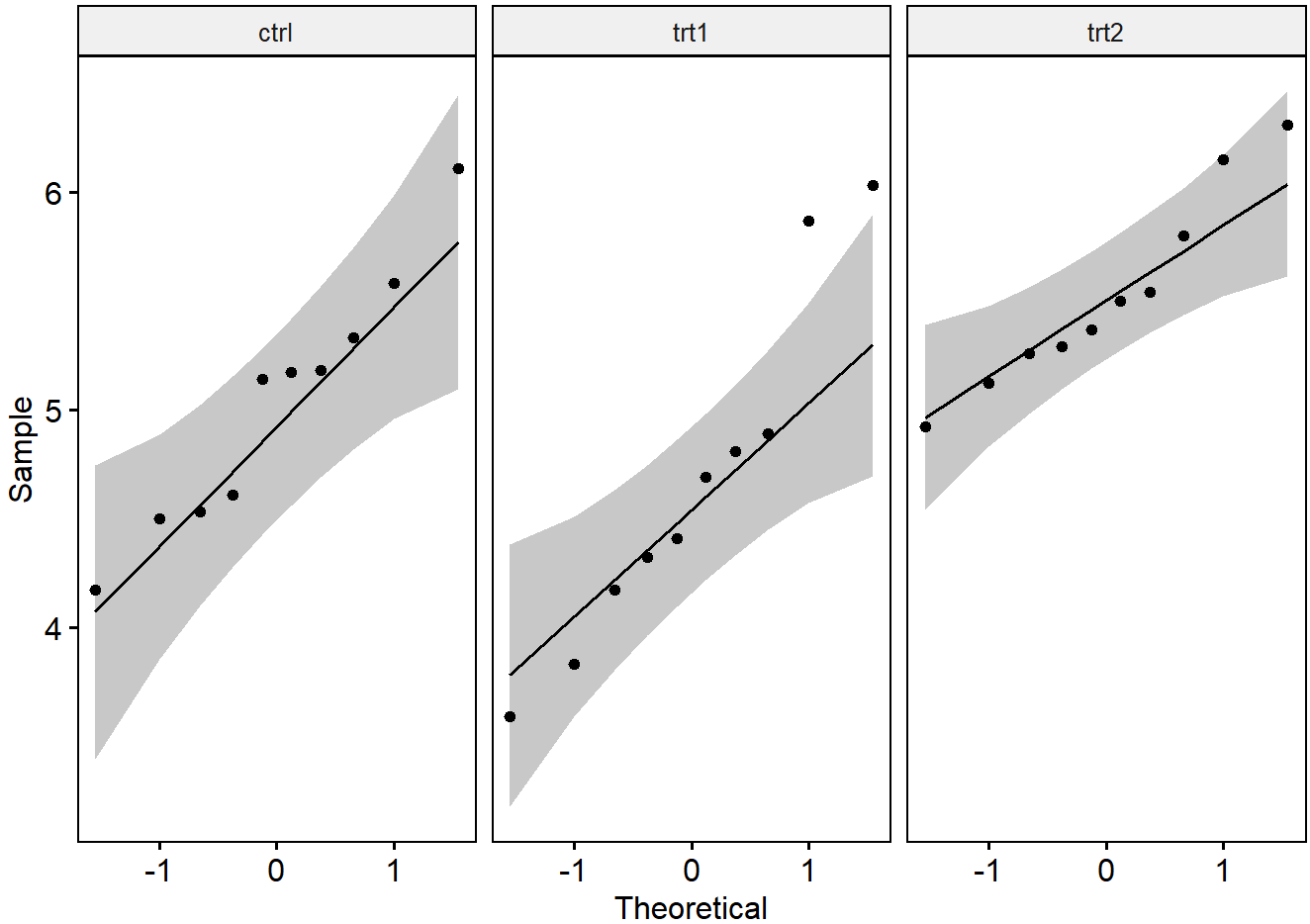
`jmv` One-way Between-Subjects ANOVA

Assumption Checks

```
PlantGrowth %>%
  group_by(group) %>%
  shapiro_test(weight)
```

```
## # A tibble: 3 × 4
##   group variable statistic     p
##   <fct> <chr>        <dbl> <dbl>
## 1 ctrl  weight       0.957 0.747
## 2 trt1  weight       0.930 0.452
## 3 trt2  weight       0.941 0.564
```

```
ggqqplot(PlantGrowth, "weight", facet.by = "group")
```




```
PlantGrowth %>%
  levene_test(weight ~ group)
```

```
## # A tibble: 1 × 4
##     df1   df2 statistic     p
##   <int> <int>     <dbl> <dbl>
## 1     2    27      1.12 0.341
```

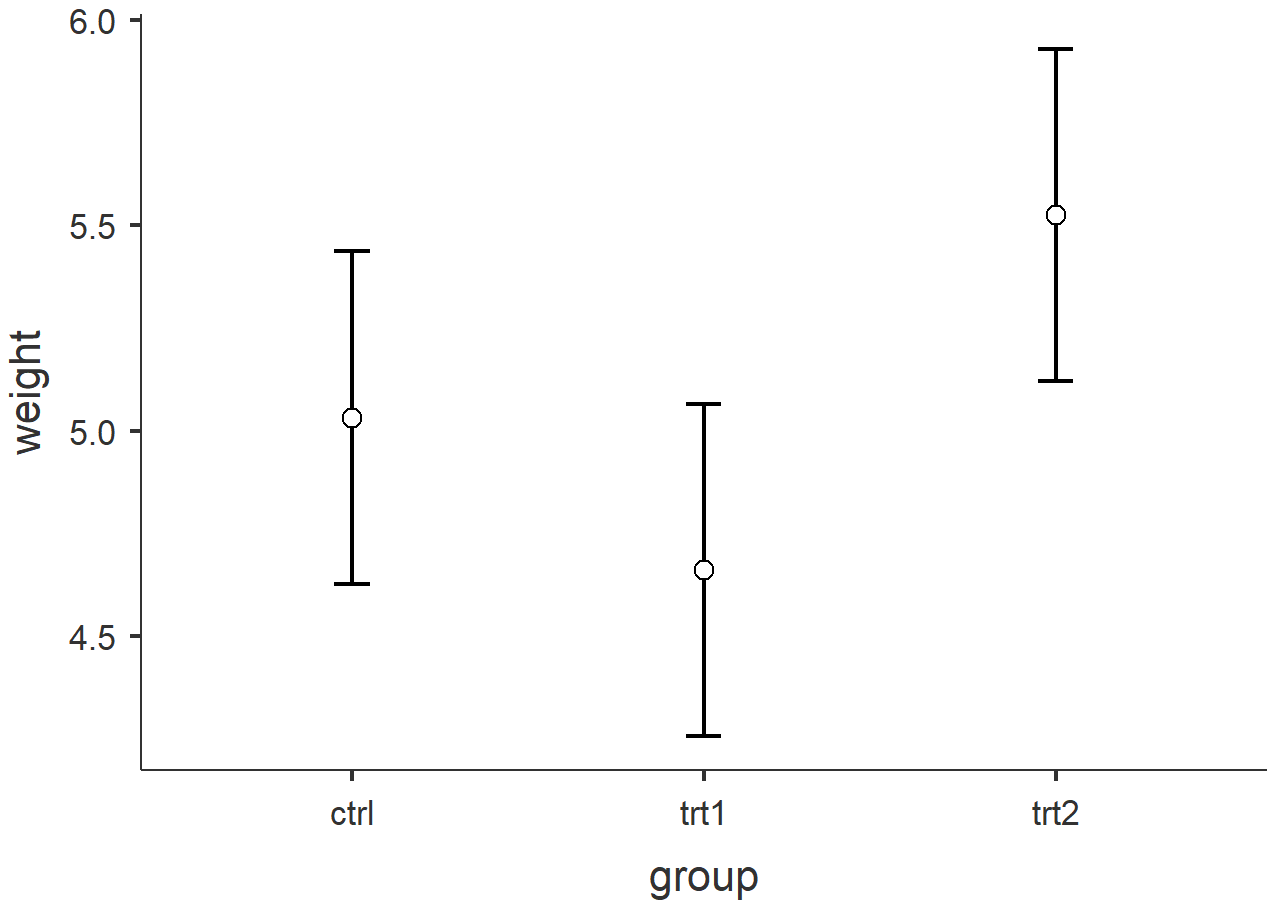
```
# Can also use the `norm = ` and `homo = ` arguments of ANOVA() to assess
# Normality and Homogeneity from a univariate perspective
PlantGrowth %>%
  ANOVA(dep=weight,
        factors = group,
        norm = TRUE,
        homo = TRUE)
```

```
##
##  ANOVA
##
##  ANOVA - weight
##  ─────────────────────────────────────────────────────────────────────────────
##                  Sum of Squares    df    Mean Square    F           p
##  ─────────────────────────────────────────────────────────────────────────────
##    group              3.766340     2      1.8831700    4.846088    0.0159100
##    Residuals         10.492090    27      0.3885959
##  ─────────────────────────────────────────────────────────────────────────────
##
##
##  ASSUMPTION CHECKS
##
##  Homogeneity of Variances Test (Levene's)
##  ────────────────────────────────────────────
##    F           df1    df2    p
##  ────────────────────────────────────────────
##    1.236963      2     27    0.3061949
##  ────────────────────────────────────────────
##
##
##  Normality Test (Shapiro-Wilk)
##  ──────────────────────────────
##    Statistic    p
##  ──────────────────────────────
##    0.9660681    0.4378986
##  ──────────────────────────────
```

Running the ANOVA

```
PlantGrowth %>%
  ANOVA(dep=weight,
        factors = group,
        effectSize = 'eta',
        norm = TRUE,
        homo = TRUE,
        postHoc = weight~group,
        postHocCorr = 'bonf',
        postHocES = 'd',
        postHocEsCi = TRUE,
        emMeans = list(list("group")),
        emmPlots = TRUE)
```

```
##
##  ANOVA
##
##  ANOVA - weight
##  ────────────────────────────────────────────────────────────────────────────────────────────────
##                  Sum of Squares    df    Mean Square    F           p           η²
##  ────────────────────────────────────────────────────────────────────────────────────────────────
##    group              3.766340     2      1.8831700    4.846088    0.0159100    0.2641483
##    Residuals         10.492090    27      0.3885959
##  ────────────────────────────────────────────────────────────────────────────────────────────────
##
##
##  ASSUMPTION CHECKS
##
##  Homogeneity of Variances Test (Levene's)
##  ──────────────────────────────────────────────
##    F           df1    df2    p
##  ──────────────────────────────────────────────
##    1.236963      2     27    0.3061949
##  ──────────────────────────────────────────────
##
##
##  Normality Test (Shapiro-Wilk)
##  ──────────────────────────────
##    Statistic    p
##  ──────────────────────────────
##    0.9660681    0.4378986
##  ──────────────────────────────
##
##
##  POST HOC TESTS
##
##  Post Hoc Comparisons - group
##  ────────────────────────────────────────────────────────────────────────────────────────────────
────────────────────────────────────────────────
##    group         group    Mean Difference    SE           df          t            p-bonferro
ni    Cohen's d     Lower         Upper
##  ────────────────────────────────────────────────────────────────────────────────────────────────
────────────────────────────────────────────────
##    ctrl     -    trt1           0.3710000    0.2787816    27.00000     1.330791       0.58316
36     0.5951477    -0.3373845     1.5276799
##             -    trt2          -0.4940000    0.2787816    27.00000    -1.771996       0.26304
50    -0.7924609    -1.7363687     0.1514470
##    trt1     -    trt2          -0.8650000    0.2787816    27.00000    -3.102787       0.01337
77    -1.3876086    -2.3836590    -0.3915582
##  ────────────────────────────────────────────────────────────────────────────────────────────────
────────────────────────────────────────────────
##    Note. Comparisons are based on estimated marginal means
```



Publishable Plot | Using `rstatix` and `ggpubr` packages

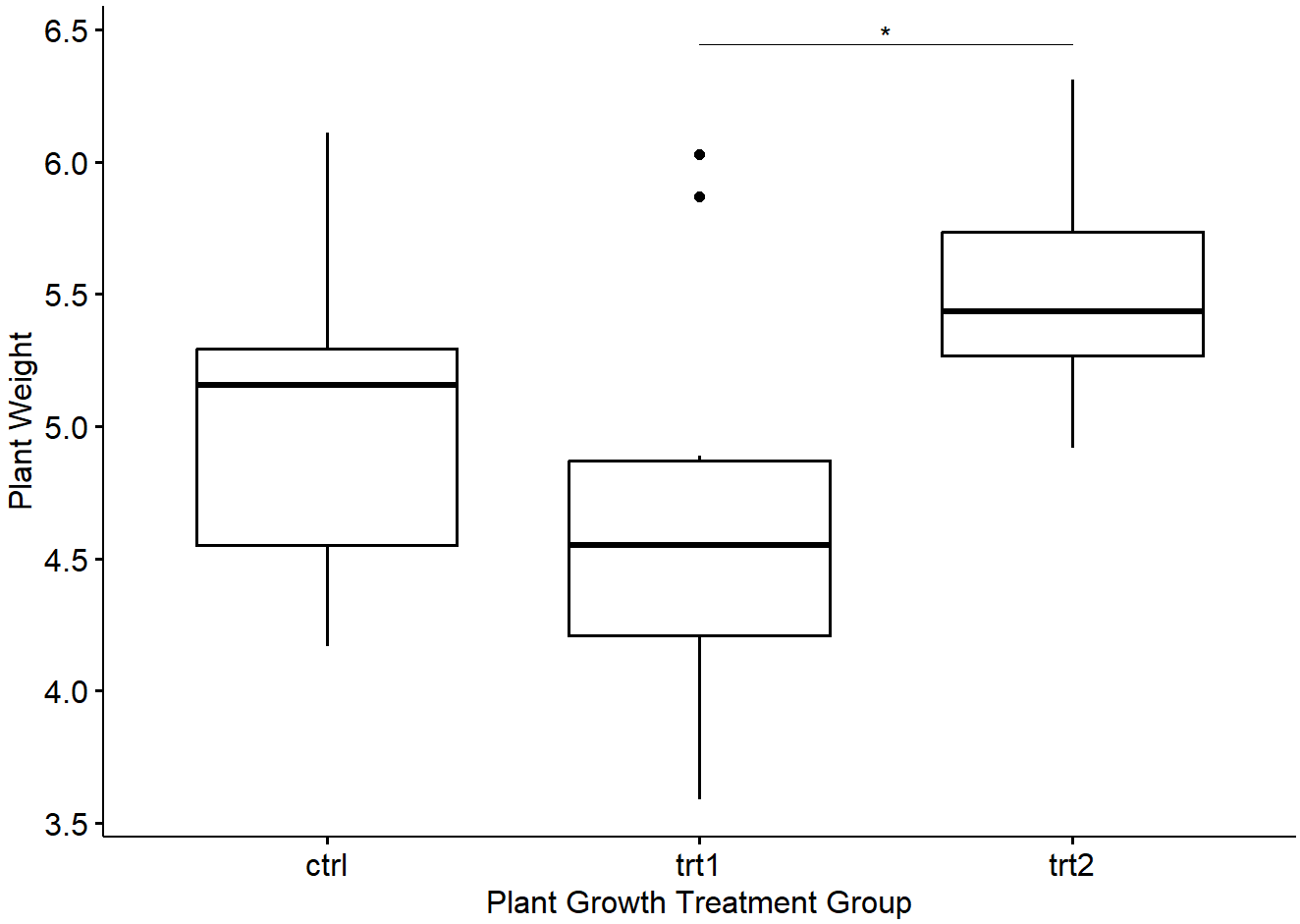
```
pwcB <- PlantGrowth %>%
  pairwise_t_test(weight ~ group,
                  p.adjust.method = "bonferroni") %>%
  add_xy_position(x = "group")

pwcB
```

```
## # A tibble: 3 × 13
##   .y.    group1 group2    n1    n2       p p.signif  p.adj p.adj.signif
##   <chr>  <chr>  <chr>  <int> <int>   <dbl> <chr>     <dbl> <chr>
## 1 weight ctrl   trt1      10    10 0.194   ns       0.583  ns
## 2 weight ctrl   trt2      10    10 0.0877  ns       0.263  ns
## 3 weight trt1   trt2      10    10 0.00446 **       0.0134 *
## # ℹ 4 more variables: y.position <dbl>, groups <named list>, xmin <dbl>,
## #   xmax <dbl>
```

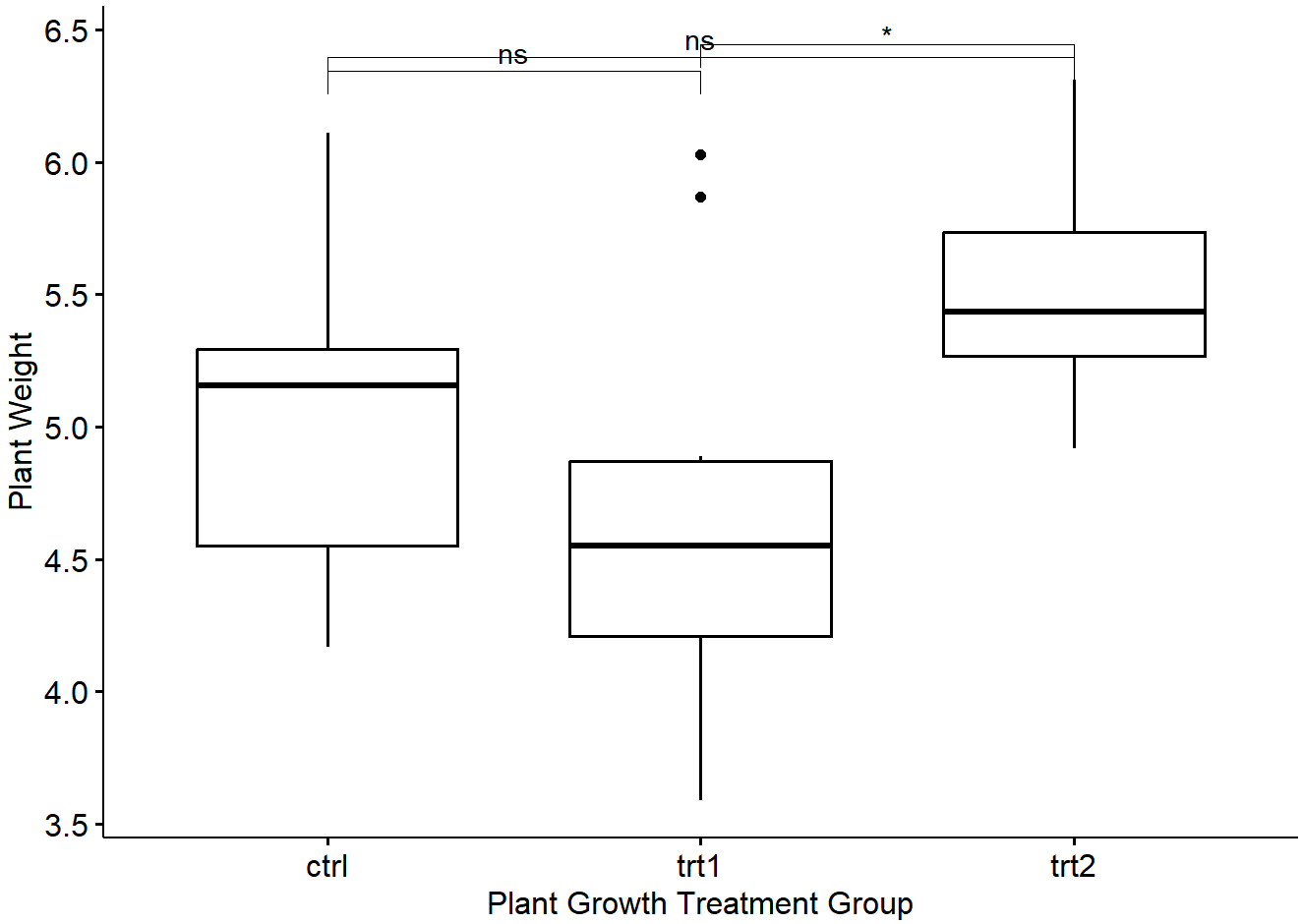
```
pwcT <- PlantGrowth %>%
  tukey_hsd(weight ~ group) %>%
  add_xy_position(x = "group")
pwcT
```

```
## # A tibble: 3 × 13
##   term  group1 group2 null.value estimate conf.low conf.high p.adj p.adj.signif
##   <chr> <chr>  <chr>       <dbl>    <dbl>    <dbl>     <dbl> <dbl> <chr>
## 1 group ctrl   trt1            0   -0.371   -1.06      0.320 0.391 ns
## 2 group ctrl   trt2            0    0.494   -0.197     1.19  0.198 ns
## 3 group trt1   trt2            0    0.865    0.174     1.56  0.012 *
## # ℹ 4 more variables: y.position <dbl>, groups <named list>, xmin <dbl>,
## #   xmax <dbl>
```

```
ggboxplot(PlantGrowth,
          x = "group",
          y = "weight") +
  labs(x = "Plant Growth Treatment Group",
       y = "Plant Weight"
  ) +
  stat_pvalue_manual(pwcB, hide.ns = TRUE, tip.length = 0)
```



```
ggboxplot(PlantGrowth,
          x = "group",
          y = "weight") +
  labs(x = "Plant Growth Treatment Group",
       y = "Plant Weight"
  ) +
  stat_pvalue_manual(pwcT, hide.ns = FALSE, tip.lenth = 0)
```



APA Style Results Section

Statistical Interpretation:

A between-subjects one-way ANOVA was performed to determine the relationship between plant growth (i.e., weight) and nutrient treatments (e.g., control, calcium, and nitrogen). The mean and standard deviations are presented in Table 1. The ANOVA was statistically significant at the .05 alpha level, $F(2, 27) = 4.85$, $p = .016$, $\eta^2 = .264$. A post hoc test was conducted using the Bonferroni Correction that indicated that the mean weight of plants treated with Nitrogen were statistically significantly higher than the mean weight of plants treated with Calcium ($p = .01$, Cohen's $d = -1.388$, 95% CI [-2.384, -0.392]). However, the mean weight of plants treated with the control was not statistically significantly different compared to the mean weight of plants treated with Nitrogen ($p = .58$, Cohen's $d = 0.595$, 95% CI [-0.337, 1.528]) or to the mean weight of plants treated with Calcium ($p = .26$, Cohen's $d = -0.793$, 95% CI [-1.736, 0.152]).